Design and Implementation of a Real-Time Geolocation Tracking System

Team & Project Guide

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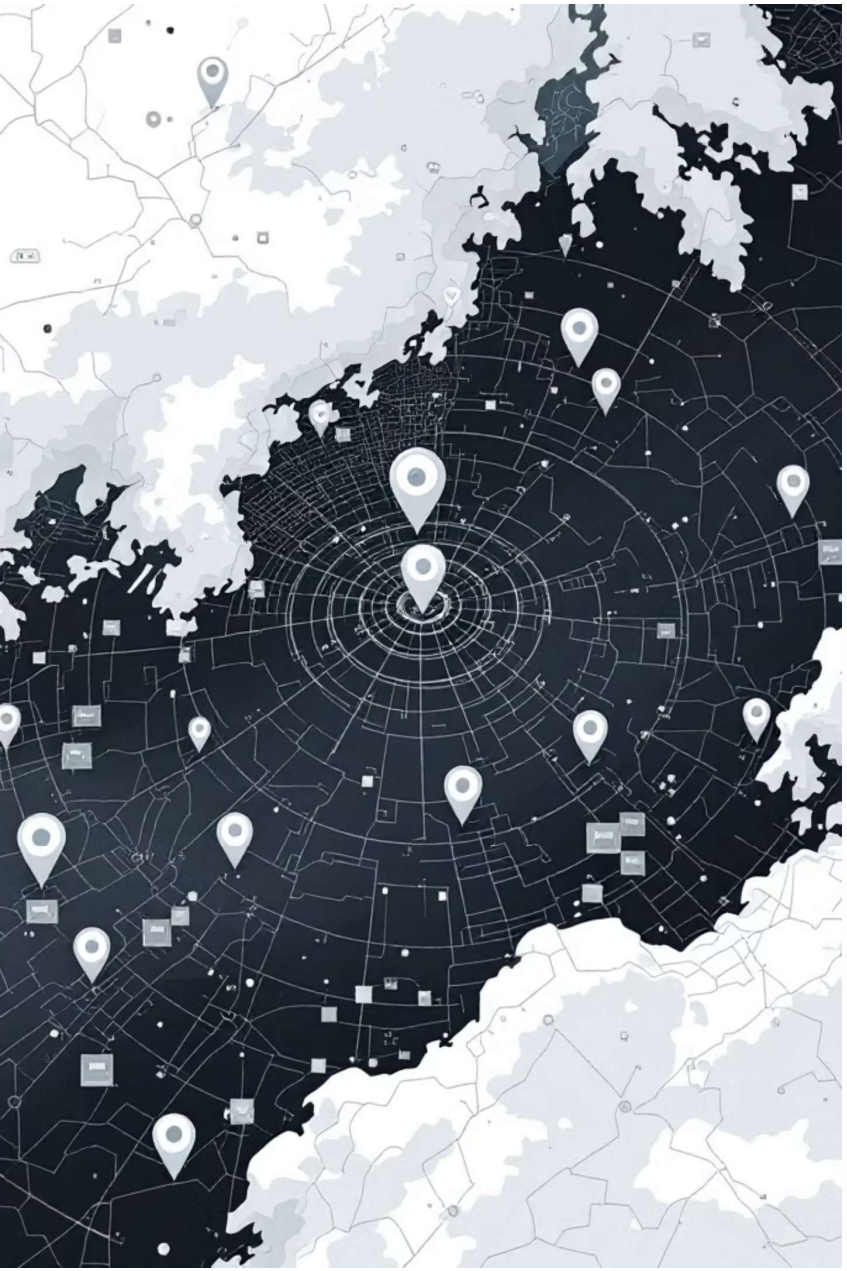

Introduction

Object Detection using Geo-Location Tracking is a system that can **identify an object and track its location at the same time**.

It uses a camera or sensor to detect the object and uses GPS or Wi-Fi signals to find its exact position.

This system helps in **tracking people, vehicles, or important items** in real time.

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Existing Limitations

Localization Gaps

GPS unreliable indoors; coarse accuracy outdoors

Detection Errors

False positives/negatives under varied lighting and occlusion

Latency

Slow processing leads to stale location data

Scalability

High compute cost for large areas and many cameras

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Problem Identification

1. Need accurate indoor + outdoor positioning
2. Reliable object association to coordinates
3. Low-latency alerts for moving objects
4. Cost-effective, scalable architecture

Design must bridge sensing, localization, and tracking pipelines.

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Proposed System Overview

- wifi-routers
- **Edge Detection**
- **Localization Fusion**
- **Central Tracking**

Key pillars: wifi-routers, Edge detection, Localization fusion, Real-time tracking, User interfaces

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Wi- Fi Based Indoor Localization

Approach

- RSSI + triangulation
- Wi-Fi RTT for sub-meter accuracy
- Calibration using labeled ground truth

Benefits

- Works with existing infrastructure
- Scalable to large buildings

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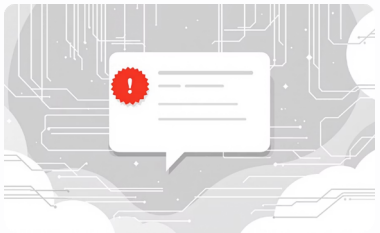
Real- Time Monitoring System

Live Map

Current positions, movement trails, heatmap



Alerts & Rules

Geofence, speed, unauthorized access

Playback

Reconstruct events; exportable logs

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Advantages

Higher Accuracy

Sub-meter indoor localization fused with detection

Low Latency

Edge inference + efficient data pipelines

Scalable

Modular architecture—add sensors & zones easily

Privacy- First

On-device anonymization; configurable retention

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Conclusion

In conclusion, the proposed system can detect objects and track their location in real time.

By using GPS for outdoor tracking and Wi-Fi for indoor tracking, it provides better accuracy and continuous monitoring.

This system improves security, safety, and easy tracking of important objects.

Thank You